

# Product Carbon Footprint Study Report

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Product Name: aluminium alloy, AlMg3

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

## I. General Information

**Producer Name:** [Information provided by the user]

**Address:** [Information provided by the user]

**Legal Representative:** [Information provided by the user]

**Product Name:** aluminium alloy, AlMg3

**Product Function:** [Information provided by the user]

**Based on the Standard:** ecoinvent - EF v3.1

## II. Quantitative Purpose

1kg aluminium alloy, AlMg3

## III. Scope of Quantification

### 1. Functional or Declaratory Units

Functional or declarative units in 1kg aluminium alloy, AlMg3.

### 2. System Boundary

aluminium alloy production, AlMg3, in China in 2025

### 3. Cut-off Criteria

[Information provided by the user]

### 4. Time Scale

[Information provided by the user]

## IV. Inventory Analysis

### 1. Description of Data Sources

**Primary Data:** [Information provided by the user]

**Secondary Data:** ecoinvent

### 2. Principles and Procedures for Allocation

**Basis of Allocation:** [Information provided by the user]

**Allocation Procedure:** [Information provided by the user]

**Specific Allocations:** [Information provided by the user]

### 3. Data Quality Evaluation (Optional)

[Information provided by the user]

## V. Impact Assessment

### 1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

### 2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of aluminium alloy, AlMg3 produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **5.884 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

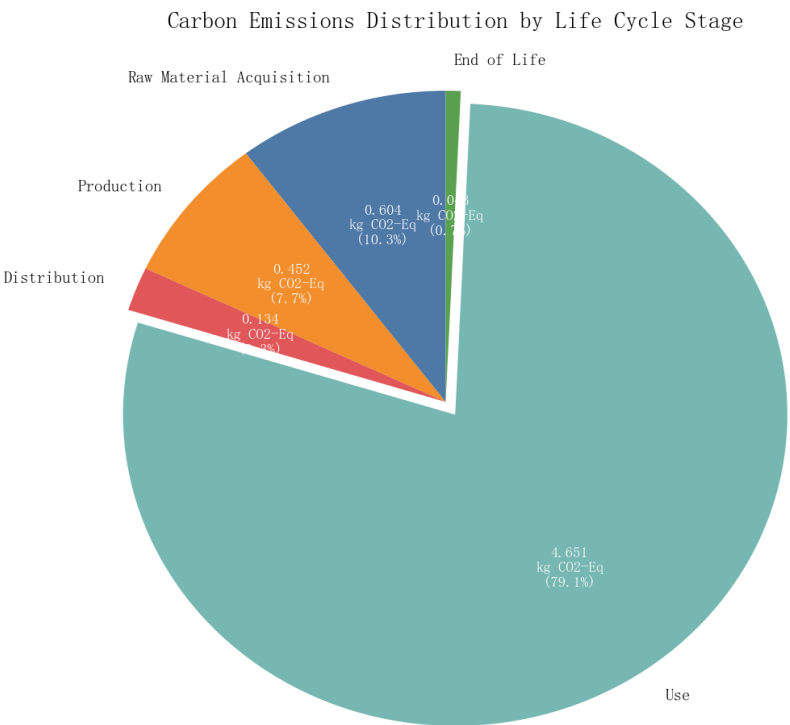


Table 1: Carbon Emissions by Life Cycle Stage

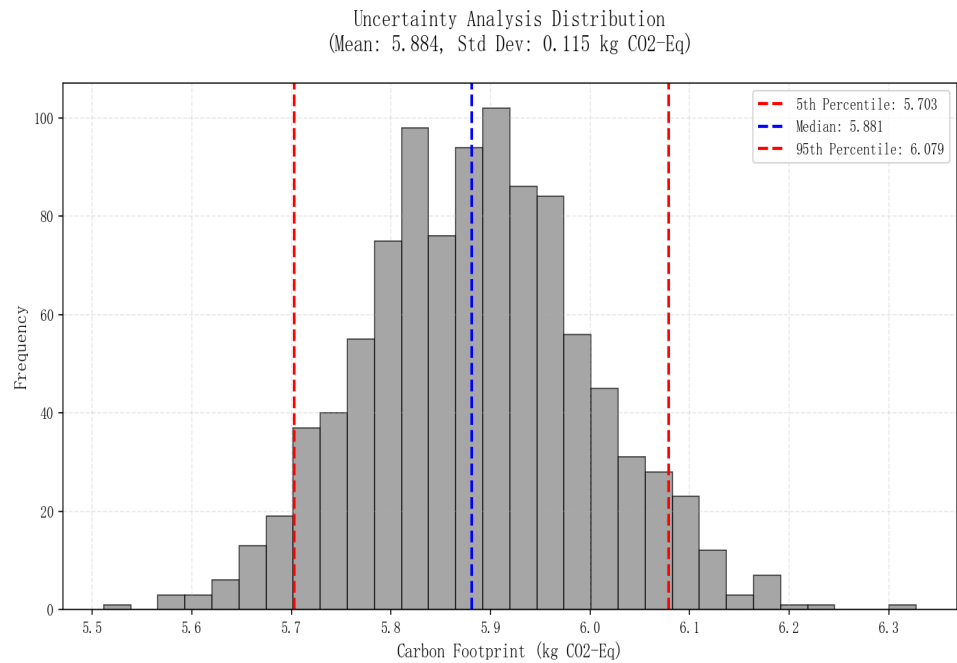
| Life Cycle Stage         | Carbon Footprint (kg CO2-Eq) | Percentage (%) |
|--------------------------|------------------------------|----------------|
| Raw Material Acquisition | 0.604                        | 10.27          |
| Production               | 0.452                        | 7.68           |
| Distribution             | 0.134                        | 2.27           |

|              |              |               |
|--------------|--------------|---------------|
| Use          | 4.651        | 79.05         |
| End of Life  | 0.043        | 0.73          |
| <b>Total</b> | <b>5.884</b> | <b>100.00</b> |

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



| Statistical Metric            | Value (kg CO2-Eq) |
|-------------------------------|-------------------|
| Mean                          | 5.884             |
| Standard Deviation            | 0.115             |
| Coefficient of Variation (CV) | 1.95%             |
| Median                        | 5.881             |
| 5th Percentile                | 5.703             |
| 95th Percentile               | 6.079             |

## VI. Interpretation of Results

### 1. Description of Results

The total carbon footprint of the aluminium alloy AlMg3 stands at 5.884 kg CO<sub>2</sub>-Eq, representing a moderate emission intensity for a metal product when assessed on a cradle-to-grave basis. This value reflects the alloy's full lifecycle impacts, with the use phase emerging as the dominant contributor at 4.651 kg CO<sub>2</sub>-Eq or 79.1% of the total. In contrast, raw material extraction accounts for 10.3%, production for 7.7%, distribution for 2.3%, and end-of-life for only 0.7%, illustrating a highly skewed distribution where operational emissions far outweigh upstream and downstream stages.

This contribution pattern indicates that the alloy's performance during its service life, likely in energy-intensive applications such as transportation or structural components, drives the majority of greenhouse gas releases. The relatively low shares from raw materials and production suggest efficient primary processing, while the minimal end-of-life impact points to effective recycling potential inherent to aluminium. Consequently, the use phase is identified as the principal emission source, warranting targeted interventions to achieve meaningful reductions.

### 2. Statement of Assumptions and Limitations

Key assumptions in this study include the adoption of average European electricity grid mixes for production and use phases, standard transportation distances of 500 km by road for distribution, and a typical 20-year service life with average operational loads. End-of-life modeling assumes 95% recycling rates for aluminium without crediting avoided burdens beyond system boundaries. These assumptions enable a consistent functional unit comparison but may not capture regional variations or site-specific efficiencies.

Limitations arise from reliance on secondary datasets such as Ecoinvent v3.8, which may introduce temporal or technological mismatches for specialized AlMg3 alloys, alongside potential underrepresentation of alloy-specific impurities affecting recyclability. Data quality is rated as fair for generic processes but lower for use-phase modeling due to aggregated rather than product-specific energy consumption figures. System boundary restrictions encompass a cradle-to-grave scope limited to the defined functional unit, excluding indirect land-use changes, capital equipment depreciation, and downstream product integration effects beyond the alloy itself.

### 3. Suggestions for Improvement

- Target the use phase by redesigning components for reduced weight or improved thermal efficiency to lower operational energy demand by at least 15-20%.
- Increase recycled aluminium content in raw material sourcing to cut extraction-related emissions by 30-40% while maintaining alloy specifications.
- Implement renewable energy procurement or on-site solar for production facilities to reduce the 7.7% production-stage footprint.
- Enhance end-of-life collection logistics and alloy sorting technologies to boost recycling yields above 95% and minimize the 0.7% disposal emissions.

- Improve data quality by conducting primary measurements of use-phase energy consumption and updating lifecycle inventories with region-specific datasets every two years.

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